

Sunghyun Park

 edipark |  Sunghyun Park |  Youtube |  edi_park@yonsei.ac.kr |  +82-10-2403-1695

Research Interest

Robotics, Learning-Based Control, Locomotion, Manipulation, Whole-Body Control, Optimal Control, Cross-Embodiment

Education

Yonsei University, Seoul, Republic of Korea

Mar 2021 - Present

B.E. in Mechanical Engineering

- Expected Graduation : Feb 2027
- Cumulative GPA: 4.11/4.3 ($\approx$ 3.92/4.0)
- Relevant Courses: Reinforcement Learning, Mechatronics, Mechanical System Control, Engineering Numerical Analysis
- 18-month absence due to military service as a KATUSA(Korean Augmented to the US Army) Medic

Research Experience

Machine Learning and Control Systems Laboratory, Yonsei University

Nov 2025 - Present

Undergraduate Researcher, (Supervisor : Prof. Jongeun Choi)

- Implemented a VR-based teleoperation system for RB-Y1
- Implemented VLA-based manipulation pipeline of RB-Y1
- **Current Status** : Developing Multi-Humanoid benchmark for visuomotor multi agent policy

Dynamic Robotic Systems Laboratory, Seoul Nat'l University

Jul 2025 – Sep 2025

Undergraduate Intern, (Supervisor : Prof. Jaeheung Park)

- Developed a reinforcement learning framework for humanoid control with motion retargeting
- Conducted research on whole-body control of humanoid robots

Selected Projects

UMMA: University Mobile Management Agent

[Project Page](#)

Mar 2026 - Jun 2026

- Creative Product Design, Department of Mechanical Engineering, Yonsei University, 2026 Spring
- Autonomous building management system
- Keywords : ROS2, YOLO, SLAM, Hardware Design

SOLO : State estimation with Only Leg Observation

[Github](#)

Dec 2025 - Apr 2026

- Deployed to **DEXTRA**
- State Estimation with leg observation history
- Keywords : Locomotion, State estimator

PALMA : A robotic hand project

[Link to Demo](#)

Dec 2025 - Feb 2026

- Printed and assembled from scratch
- Vision based teleoperation
- Keywords : Computer Vision, Tendon Actuation

DEXTRA : A low cost humanoid project

[Github](#)

Aug 2025 - Jan 2026

- Designed, printed, and assembled from scratch
- Deepmimic based bipedal walking motion learning
- Keywords : Motion Retarget, RL

Publications & Working Papers

- Hyeonjung Kim, **Sunghyun Park**, and Hyungjun Jeon, "SOLO: State estimation with Only Leg Observation for IMU-Free Bipedal Locomotion," (in preparation).

Selected Awards & Honors

Grand Prize, Course Outcome-based Idea Competition , Yonsei University	2026 Summer
• UMMA: University Mobile Management Agent • Qualified for the National Championship	
Academic Excellence Scholarship , Yonsei University	2022 Spring - Present
• Ranked 1st out of 231 students in the Department of Mechanical Engineering (Fall 2024) • Selected for 5 semesters, excluding period of military service	
National Scholarship (Type I & II) , Korea Student Aid Foundation (KOSAF)	2021 Fall - Present
• Merit-based financial support awarded for maintaining excellence in undergraduate studies • Selected for 6 semesters, excluding period of military service	
3rd place, 2025 STEAMCUP IRC Turtlebot3 AutoRace	Nov 2025
• Implemented autonomous driving framework for Turtlebot3 • Operated based on : ROS2, Linux Ubuntu, Flexbe, RaspberryPi	

Extracurricular Activities and Leadership

YAI, Artificial Intelligence Club , Yonsei University	Jun 2024 - Present
• Studied basics of AIs and discussed latest robot learning papers • AI Projects : Learning Bipedal walking of DEXTRA, RL-HFT project	
RoboIn, Robotics Club , Yonsei University	Mar 2024 - Present
• Executive Staff(2024-2025), Vice President(2026) • Robot Projects : DEXTRA(humanoid), PALMA(robotic hand), Turtlebot, Manipulator • Conducted Seminars : Basics of humanoids, Robot Learning : Starting from the Reinforcement Learning	

Skills

Programming Languages	Python, C, C++, MATLAB
Frameworks/Tools	ROS2, Docker, Isaac Gym, Isaac Lab, MuJoCo, Fusion360, Inventor, CREO, ANSYS
Hardware	RaspberryPi, myRIO, Arduino, Jetson series, Dynamixel series, Turtlebots
Languages	Korean(Native), English(Fluent, iBT TOEFL 5.5(110))